Chapter 4: Vacuum Dynamics, State Transitions, and Matter-Antimatter Asymmetry

4.1 The Vacuum as a Relaxed Continuum

The BMI framework defines the vacuum state as the **baseline, fully relaxed configuration of the bimodal manifold**. When no topological twists are present, the overlapping branes (σ_A and σ_B) oscillate in global Metric Equilibrium (Δ). Matter is not an object placed *into* the vacuum; it is a high-tension, phase-locked deformation *of* the vacuum. Propagation is the transmission of localized manifold tension, and the photon is the minimal kinetic wave packet representing the manifold's elastic transition.

4.2 Annihilation as Topological Phase Cancellation

Annihilation is the structural phase-cancellation of localized manifold tension. When a matter harmonic and an antimatter harmonic intersect, their geometric distortions invert, releasing the "frozen" Bridge Stress (τ) back into the global manifold as radiative energy. This confirms that matter-to-light conversion is a macroscopic manifestation of geometric relaxation to the baseline equilibrium.

4.3 Geometric Hysteresis and Baryon Asymmetry

The Baryon Asymmetry is a result of **Geometric Hysteresis** during the initial manifold collision. At the "Cradle" epoch, the Bridge Stress (τ) was at its maximum. The energy required to create a positive topological twist was subtly lower than that required for antimatter, due to a chiral bias in the manifold torsion:

$$\Delta E_{matter} = E_0 \cdot (1 - \tau_{global}) < \Delta E_{antimatter} = E_0 \cdot (1 + \tau_{global})$$

As the Bridge Stress dissipated, the "matter-favorable" configurations were frozen into the topology. The observable matter is the geometric survivor of this relaxation event, representing the stable ground-state of a manifold that once possessed sufficient energy density to break symmetry.